

Teo extension entera ideal primo maximal iff maximal

Teorema 1. *Sea $A \subset B$ una extensión entera de anillos y sea $\mathfrak{p} \subset B$ un ideal primo. Entonces, $\mathfrak{p}$ es maximal $\iff \mathfrak{p} \cap A$ es maximal.*

Demostración: Por el ejer-extension-entera-cociente/??, $A/\mathfrak{p} \cap A \subset B/\mathfrak{p}$ es una extensión entera de anillos. Además, como $\mathfrak{p}$ es primo, $\mathfrak{p} \cap A$ es primo, luego por la prop-ideal-primo-iff-cociente-di-integridad/Proposición 1, $B/\mathfrak{p}$ y $A/\mathfrak{p} \cap A$ son dominios de integridad.

Entonces, por el teo-extension-entera-dominios-integridad-imp-cuerpo-iff-cuerpo/Teorema 1, $A/\mathfrak{p} \cap A$ es un cuerpo $\iff B/\mathfrak{p}$ es un cuerpo. Luego, por la prop-ideal-maximal-iff-cociente-cuerpo/, $\mathfrak{p} \cap A$ es maximal $\iff \mathfrak{p}$ es maximal. ■

Referenciado en

- teo-extension-entera-ideal-primo-imp-exists-ideal-primo-contrae

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.